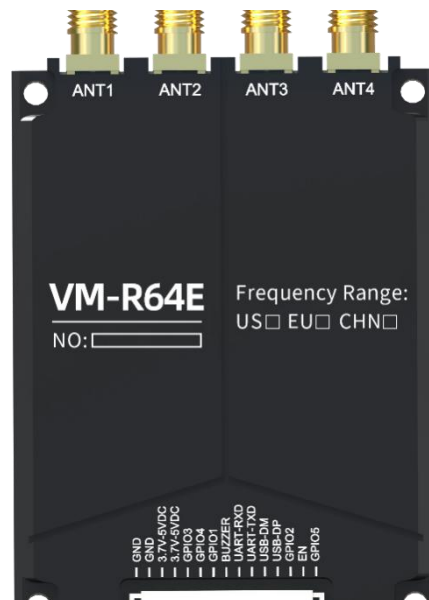


# VM-R64E



## Four Port E710 Module Product Specification



## Product Features:



The VM-R64E four-channel module is designed based on the RF chip E710 of the American company Impinj. It adopts the "adaptive dynamic Q anti-collision algorithm" and has excellent multi-tag group reading and long-distance identification performance. The peak speed of inventory tags is greater than 850/Seconds, the module has 4 RF SMA antenna connection ports, which can connect 4 antennas at the same time to achieve time-sharing cycle inventory. The output power range is 0~33dBm. When using a 9dBi circularly polarized antenna, the stable identification distance is greater than 25 meters.

It can be used in fixed RFID reader/writer integration, RFID access control channels, smart RFID cabinets and other applications that require multi-antenna identification and reading.

➤ **High performance multi-label recognition algorithm**

Based on the American Impinj company's dynamic Q algorithm and a new data model, the multi-tag reading performance is significantly improved. When the number of tags increases, the performance becomes more obvious;

➤ **High power output capability**

Using a high-power PA chip, the maximum output power of the module can reach 34.5 dBm, and the PA still maintains a linear state when outputting 33dBm;

➤ **Dual-CPU-like architecture design greatly improves overall performance**

E710 chip is responsible for polling tags, and the CPU is responsible for data management. Polling tags and sending data are done in parallel and do not occupy each other's time;

➤ **Excellent long-distance recognition performance**

Ultra-low receiving sensitivity and high power output capability make it easy to identify longer-distance tags;

➤ **Antenna connection detection function**

The module's integrated antenna physical connection detection circuit can quickly and accurately detect the antenna connection status of the module port ;

➤ **RF power amplifier PA protection function**

The module integrates forward and reverse power detection. When the module RF port detects the unconnected antenna status, it starts to protect the PA from being burned out in the case of open

circuit output;

➤ **Onboard temperature sensor**

Onboard high-precision temperature sensor accurately monitors the operating temperature of the system in real time;

➤ **Outstanding thermal design**

The bottom of the module has a large area of contact with the heat sink, and all heating devices have thermal coupling interfaces with thermal conductivity structures, and use thermal conductive silicone grease with high thermal conductivity, which is non-volatile at high temperatures;

➤ **Excellent stability**

The precision CNC design of the aviation aluminum alloy shell shields electromagnetic environmental interference, wide temperature design , and extremely low temperature drift coefficient. Work 24 hours x 365 days without downtime;

## Specifications:

|                                            |                                                                                                                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purchasing code<br/>(product model)</b> | VM-R64                                                                                                                                                                        |
| <b>Performance parameters</b>              |                                                                                                                                                                               |
| Operating Voltage                          | DC 3.6V ~ 5V                                                                                                                                                                  |
| Standby operating current                  | < 50 mA (EN pin high level enable)                                                                                                                                            |
| Sleep working current                      | < 100u A (EN pin low level enable)                                                                                                                                            |
| Operating current                          | 2.1A @ 5V ( 33dBm CW Output, 25 °C ).<br>1.5 A @ 5V ( 30 dBm CW Output, 25 °C ).<br>Average operating current 1.7 A @ 5V ( 33 dBm Output, real-time inventory mode , 25 °C ). |
| Start Time                                 | < 50mS                                                                                                                                                                        |
| Operating temperature                      | -20°C ~ +70°C                                                                                                                                                                 |
| storage temperature                        | - 40 °C ~ + 85 °C                                                                                                                                                             |
| Working humidity                           | < 95% ( + 25 °C)                                                                                                                                                              |
| air interface protocol                     | EPC global UHF Class 1 Gen 2 / ISO 18000-6C                                                                                                                                   |

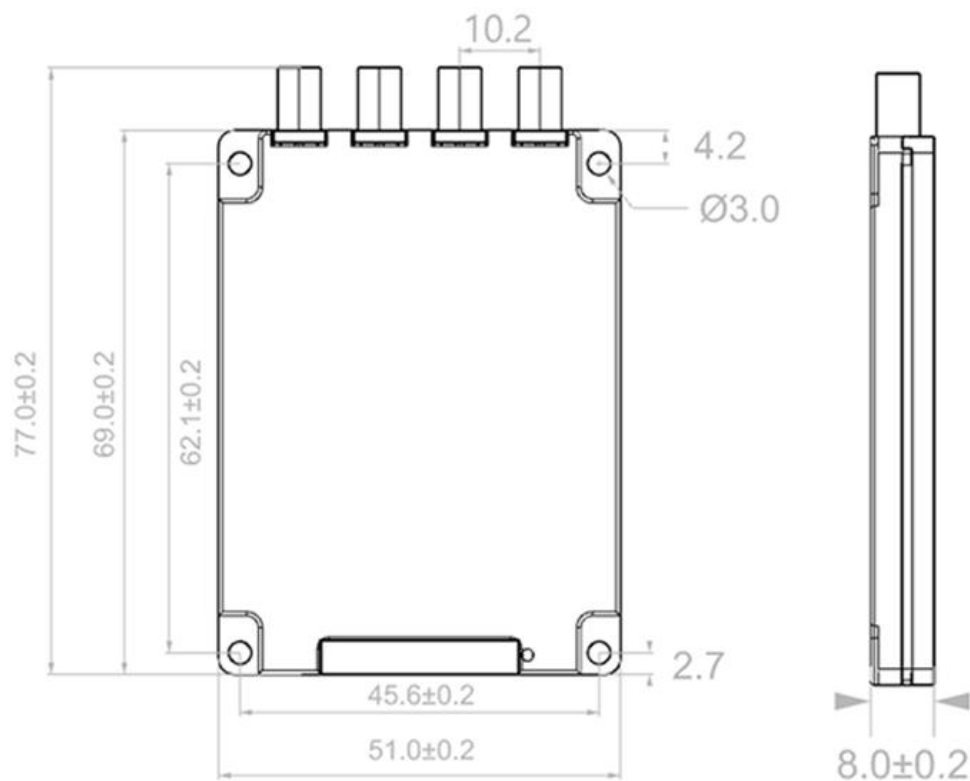
|                                |                                                                                                                                                                                              |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Working frequency range        | 902MHz ~ 928MHz (U.S. frequency band), 865MHz ~ 868MHz (European frequency band)<br>860 MHz ~ 960 MHz (customized working frequency band)                                                    |
| Work area support              | US, Canada and other regions following US FCC<br>Europe and other regions following ETSI EN 302 208 with & without LBT regulations<br>Mainland China<br>Japan<br>Korea<br>Malaysia<br>Taiwan |
| Output power range             | 1 ~ 33dBm                                                                                                                                                                                    |
| Output power accuracy          | +/- 1dB                                                                                                                                                                                      |
| Output power flatness          | +/- 0.2dB                                                                                                                                                                                    |
| Output RF connector            | SMA RF interface                                                                                                                                                                             |
| Receive sensitivity            | < -85dBm                                                                                                                                                                                     |
| Inventory tag peak speed       | > 850 pictures/second                                                                                                                                                                        |
| TagRSSI                        | support                                                                                                                                                                                      |
| Antenna connection protection  | support                                                                                                                                                                                      |
| Working temperature monitoring | support                                                                                                                                                                                      |
| Number of antenna ports        | 4 SMA male female threads                                                                                                                                                                    |
| Communication Interface        | TTL UART interface                                                                                                                                                                           |
| GPIO                           | 2 GPIO inputs, 2 GPIO outputs (3.3V TTL level)                                                                                                                                               |
| Communication baud rate        | Standard 115200 bps (optional 38400bps and 460800bps)                                                                                                                                        |
| Heat dissipation method        | Shielding shell and bottom heat sink dissipate heat. Additional heat dissipation must be considered when working for a long time above 30dBm.                                                |

## Pin definition:

| PIN | definition | illustrate                |
|-----|------------|---------------------------|
| 1   | GND        | ground                    |
| 2   | GND        |                           |
| 3   | VCC        | Power supply DC 3.6V ~ 5V |

|    |          |                                                                                                             |
|----|----------|-------------------------------------------------------------------------------------------------------------|
| 4  | VCC      |                                                                                                             |
| 5  | GPIO3    | No. 1 3.3V TTL level GPIO output                                                                            |
| 6  | GPIO4    | The second 3.3V TTL level GPIO output                                                                       |
| 7  | GPIO1    | No. 1 3.3V TTL level GPIO input                                                                             |
| 8  | BUZZER   | Buzzer control, requires external driver (internal driver optional)                                         |
| 9  | UART_RXD | UART RXD (module TTL serial port data input)                                                                |
| 10 | UART_TXD | UART TXD (module TTL serial port data output)                                                               |
| 11 | NC       | reserved                                                                                                    |
| 12 | NC       |                                                                                                             |
| 13 | GPIO2    | 2nd 3.3V TTL level GPIO input                                                                               |
| 14 | EN       | High level enable turns on the module (default internal pull-up is enabled), low level turns off the module |
| 15 | NC       | reserved                                                                                                    |

## Packaging specifications:



Module length, width, mounting holes and thickness dimensions